

Achouri Malek AI DevOps & Cloud Engineer

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PROFILE

AI DevOps & Cloud Engineer who automates how software and infrastructure are built, deployed and monitored, and who digs into production issues until they make sense. Hands-on with AWS, Terraform, Kubernetes and CI/CD in production for international clients. Now focused on making AI agents reliable in production: testable, observable and safe for real users.

TECHNICAL SKILLS

AI agents & LLMs: Claude API (Anthropic SDK), tool/function calling, prompt and conversation-flow design, JSON-schema tool inputs, conversation replay testing and agent evaluation, MCP (Model Context Protocol), deep learning (CNNs, 3D U-Net)

Cloud Platforms & Infrastructure: AWS , Azure , OpenStack

Containerization & Orchestration: Docker , Kubernetes , OpenShift, Helm, Kustomize, Container Networking

CI/CD Tools & Automation: Jenkins, GitLab CI/CD, ArgoCD, Terraform, Ansible , Bitbucket

Monitoring & Observability: Prometheus, Grafana, Alertmanager, Fluentd, ELK, OpenSearch, CloudWatch, CloudTrail

Networking & Security: HAProxy, Keepalived ,Keycloak ,NAT, VPN, VPC, SG, NACLs, AWS IAM, SSM, Ingress-nginx, cert-manager, TLS,Load Balancers, WAF, RBAC, secrets & managed identities

Testing & QA: Manual Testing , Test Automation (Postman, Selenium), Jira Xray , Gherkin (BDD)

Python & backend: Python (FastAPI, Flask, pytest, SQLAlchemy), SQL (PostgreSQL), REST APIs, Bash, PowerShell, JavaScript (Node.js, Angular), Java

PROFESSIONAL EXPERIENCE

Cloud & DevOps Engineer (Hilbert Investment Solutions), *NeoshoreTunisie*

01/2026 – Present | Tunis, Tunisia

- Architected and automated highly available, secure and scalable AWS infrastructure with Terraform (Infrastructure as Code), across separate Development, Staging and Production environments.
- Developed modular, reusable Infrastructure as Code enabling consistent multi-account and multi-region deployments, reducing configuration drift between environments.
- Industrialized secure DevSecOps CI/CD pipelines integrating automated testing, security scanning and zero-downtime continuous deployments, minimizing manual interventions and human error in production.
- Containerized and orchestrated applications with Docker and Docker Compose (OCI standards), and streamlined Linux server provisioning and hardening with Ansible, enforcing least-privilege access through AWS IAM and SSM.
- Established centralized monitoring and logging with Amazon CloudWatch and CloudTrail, enabling proactive incident detection, faster troubleshooting and full audit traceability, and defined RTO/RPO objectives from the infrastructure design stage.
- Engineered secure AWS network architectures (Site-to-Site VPN, Application Load Balancers, WAF, Route 53, Cato Networks) and led the diagnosis and resolution of critical production connectivity incidents.
- Configured SFTP infrastructure for clients in Belgium and Italy, and deployed highly available Multi-AZ Amazon RDS databases supporting business-critical operations.
- Conducted security assessments, vulnerability remediation and backup optimization, strengthening infrastructure resilience while reducing operational costs.
- Delivered and supported AWS infrastructure for international clients in the UK, France, Belgium and Italy, ensuring operational consistency and high availability across different regulatory and network contexts.

Cloud & DevOps Engineer - Design of a Distributed Private Cloud with OpenStack for Resilient Applications on Kubernetes, *KPIT Technologies*

02/2025 – 07/2025 | Sfax, Tunisia

- Architected a multi-node private cloud with OpenStack, Ceph and Kolla-Ansible, reducing deployment time by 40%.
- Cut public-cloud spend by 30–40% while maintaining 99.9% availability.
- Configured Ceph distributed storage for scalable, redundant and fault-tolerant workloads.
- Engineered segmented virtual networks (OpenStack, VXLAN, NAT, security groups) to isolate workloads.
- Made the control plane highly available with HAProxy and Keepalived.
- Provisioned VMs and networks with Terraform, reaching 100% automated deployments.
- Ran a K3s Kubernetes cluster on OpenStack VMs to orchestrate containerized applications.
- Centralized metrics and logs (Prometheus, Grafana, Fluentd, OpenSearch), shortening incident detection and response time.

Platform Engineer - Argo CD Dashboard Development, *Sofrecom Tunisia*

07/2024 – 08/2024 | Sfax, Tunisia

- Set up a K3s cluster and automated deployments with GitLab CI/CD and Argo CD, reducing deployment errors by 40%.
- Introduced a GitOps workflow that keeps cluster state in sync with Git, making releases faster and more consistent across development and testing environments.
- Built a custom Argo CD dashboard and configured alerting with Prometheus, Grafana and Alertmanager, cutting incident detection time by 50%.

Cloud & DevOps Engineer - OpenStack Cloud Deployment & Containerization, *Primatec Engineering*

06/2024 – 08/2024 | Sfax, Tunisia

- Stood up a private OpenStack cloud (Packstack) and benchmarked 3 installation methods (Packstack, Kolla-Ansible, DevStack) for performance, scalability and operational complexity.
- Containerized web applications with Docker and ran them on Minikube, streamlining local development.
- Wrote setup and configuration documentation to support reproducibility and knowledge transfer.

DevOps Engineer - Deployment Automation with Ansible, *GENIOS*

02/2022 – 05/2022 | Mahdia, Tunisia

- Automated multi-machine provisioning and rollout (Docker, DNS, critical services) with Ansible playbooks.
- Reduced manual intervention through proactive maintenance and streamlined operations.
- Standardized configuration management for repeatable deployments across multiple servers.

FEATURED PROJECTS

Conversational AI Agent Service

09/2026

- Built a customer-support AI agent (Python, FastAPI, Claude API) with tool calling (order lookup, escalation to a human) and multi-turn conversation state stored in PostgreSQL; every tool call is validated against a JSON schema.
- Designed a test suite that replays realistic multi-turn conversations from JSON fixtures and verifies outcomes (tools called, arguments, final database state) instead of exact wording; failing tests print the full transcript for debugging.
- Added a deterministic mock LLM so all 23 tests run in CI in about 2 seconds with no API key; the same scenarios also run against the real model or a live container.
- CI/CD with GitHub Actions: tests replayed against the built container, Trivy gate (HIGH/CRITICAL), push to GHCR with no stored credentials (OIDC-signed provenance); non-root multi-stage Docker image; Prometheus metrics for requests, tool success/failure and latency.

Cloud-native GitOps Platform on AKS

07/2026

- Provisioned an AKS cluster, ACR and Key Vault with Terraform, using Azure Workload Identity for secretless authentication.
- Built a GitHub Actions pipeline with multi-stage Docker builds, Trivy scanning and OIDC-based push to a private registry.
- Delivered with Argo CD (app-of-apps): Helm releases, ingress-nginx and cert-manager with automatic Let's Encrypt TLS.
- Deployed a Python MCP server on Kubernetes that exposes cluster-introspection tools to an LLM agent under least-privilege RBAC.

OTHER PROJECTS

TunisiePara Web Application Testing

10/2025

- Designed and executed functional and user acceptance tests (UAT) for an e-commerce parapharmacy platform.
- Created detailed User Stories, Test Cases, and Gherkin scenarios covering critical business workflows.
- Validated post-deployment application behavior to ensure functionality, performance, and reliability.

Terraform & Azure DevOps Automation

09/2025

- Automated enterprise-grade infrastructure using Terraform on Microsoft Azure.
- Implemented CI/CD pipelines with Azure DevOps for efficient environment provisioning and deployment.
- Designed reusable Terraform modules and managed multi-environment scalability.

CI/CD Automation with AWS Step Functions

09/2024 – 01/2025

- Automated workflows with AWS Step Functions, Lambda, and CloudFormation.
- Integrated CI/CD with Boto3 for seamless deployments.

EDUCATION

Telecommunications Engineering, Cloud infrastructure and networking,

2022 – 06/2025 | Sfax, Tunisia

National School of Electronics and Telecommunications of Sfax (ENET'Com)

Bachelor's degree in computer science and communication

2019 – 2022 | Mahdia, Tunisia

technologies, Higher Institute of Applied Sciences and Technology of Mahdia

CERTIFICATIONS

- AI Infrastructure: LLM-D, vLLM and GPUs [🔗](#)
- Fundamentals of MLOps [🔗](#)
- DevSecOps - Kubernetes DevOps & Security [🔗](#)
- OpenShift 4 [🔗](#)
- Terraform Basics Training Course [🔗](#)
- GitLab CI/CD: Architecting, Deploying, and Optimizing Pipelines [🔗](#)
- AWS Cloud Practitioner (CLF-C02) [🔗](#)
- Artificial Intelligence on Microsoft Azure [🔗](#)
- Introduction to Containers with Docker, Kubernetes, and Open Shift
- GitOps with ArgoCD [🔗](#)
- Introduction to AWS Data Pipeline [🔗](#)
- Getting Started with Jenkins [🔗](#)

LANGUAGES

Arabic — Native/Bilingual

English — Proficient

French — Fluent

AWARDS

1st Place - Green Tech Hackathon,

17/10/2024

WE-SPICE Program – Innovation training (TU Chemnitz & DRÄXLMAIER)

VOLUNTEER EXPERIENCE

Microsoft Tech Club - ENET'Com, General Secretary

09/2023 – 2026 | Sfax, Tunisia

IEEE ENET'Com, Active member

09/2022 – 2026 | Sfax, Tunisie

- Organized an event on artificial intelligence and agriculture at ISSAT Mahdia.